# Saurabh Takle

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## Professional summary

Data Engineer with 3+ years of experience building ETL pipelines, cloud data solutions, and analytics platforms across AWS, Azure, and GCP/BigQuery, with hands-on NLP/LLM work (RAG, embeddings, vector search) and production MLOps experience (model registry, drift detection, CI/CD for ML). Applying that pipeline-and-production-ML background to data science work spanning LLMs, vector retrieval, and analytics automation.

## Core competencies

Python & SQL • NLP & LLMs (Transformers, RAG) • Vector Embeddings & Retrieval • GCP & BigQuery •
MLOps & Model Monitoring • ETL & Data Pipelines • Cloud (AWS, Azure, GCP)

## Work experience

**The Home Depot** — Jan 2025 - Jul 2026
*Data Engineer* — *Atlanta, GA*

- Used Google Cloud Storage to ingest and curate daily sales, B2B order, and direct online order logs for handoff to data science and analytics teams, and automated BigQuery-based consolidated reporting on daily and monthly cycles, saving roughly 2 hours per reporting cycle
- Architected a Medallion-based data lake on AWS S3, ingesting raw supply chain data into a bronze layer, cleansing and deduplicating via Glue into a silver layer, and aggregating business-ready gold tables for downstream analytics
- Implemented data quality checks and pipeline monitoring to validate incoming datasets, improving the accuracy, consistency, and reliability of downstream reporting workflows across teams
- Developed scalable data models for inventory and sales datasets, improving consistency and usability across reporting and analytics workflows

**AVSI Systems LLC** — Aug 2023 - Dec 2024
*Data Engineer* — *Jersey City, NJ*

- Built data pipelines using Hugging Face and AWS Lambda to feed structured conversational data into an NLP model, supporting 92% accuracy on customer interaction classification
- Designed a centralized model registry using MLflow to track artifacts, parameters, and versioning across development and production environments
- Implemented data and concept drift detection using Evidently AI, alerting the engineering team when input feature distributions shifted significantly from training data
- Designed and automated end-to-end ML pipelines using Airflow and Kubernetes, reducing manual retraining effort by 40%

## Projects

**LLM-Powered Data Assistant**
Built an LLM-powered data assistant using LangChain and RAG to answer natural-language questions over curated enterprise datasets and technical documentation; developed ingestion and embedding pipelines in Azure Databricks to preprocess documents, generate vector embeddings, and retrieve relevant context for grounded LLM responses.
*Azure Databricks, RAG, LangChain*

**End-to-End Data Pipeline**
Built a real-time data pipeline using Apache Kafka, Spark Streaming, and Snowflake to ingest, process, and store live financial transaction data, with fault-tolerant processing and optimized Spark/Snowflake transformations reducing end-to-end latency.
*Kafka, Apache Spark, Snowflake*

**Osteoarthritis Detection from Hand X-Rays** *MS Coursework*
Led a 3-person team building and comparing a classical image-processing pipeline (MATLAB) and a deep-learning pipeline (YOLO, ResNet, TensorFlow, Keras) to grade osteoarthritis severity from annotated hand X-ray images, delivering the top-performing model in a class of 40+ students and winning best deep-learning model and best classical algorithm.
*TensorFlow, Keras, YOLO, ResNet, MATLAB*

## Education

**M.S. in Data Science** *Pace University, New York City, USA* — 2023
**B.E. in Information Technology** *Vidyalankar Institute of Technology, Mumbai, IN* — 2021

## Certifications

**Machine Learning** *Andrew Ng / DeepLearning.AI*

## Skills

**Languages:** Python, SQL, Scala, Java
**NLP & ML:** Hugging Face Transformers, NLP, LangChain, RAG, TensorFlow, PyTorch
**Cloud & Data Platform:** GCP (GCS, BigQuery), AWS (S3, Glue, Lambda, Redshift, SageMaker), Azure (Databricks, Synapse)
**Data Engineering:** Apache Spark, PySpark, Airflow, ETL, Data Modeling, Kafka
**MLOps:** MLflow, Evidently AI, Prometheus, Grafana, Docker, Kubernetes, CI/CD